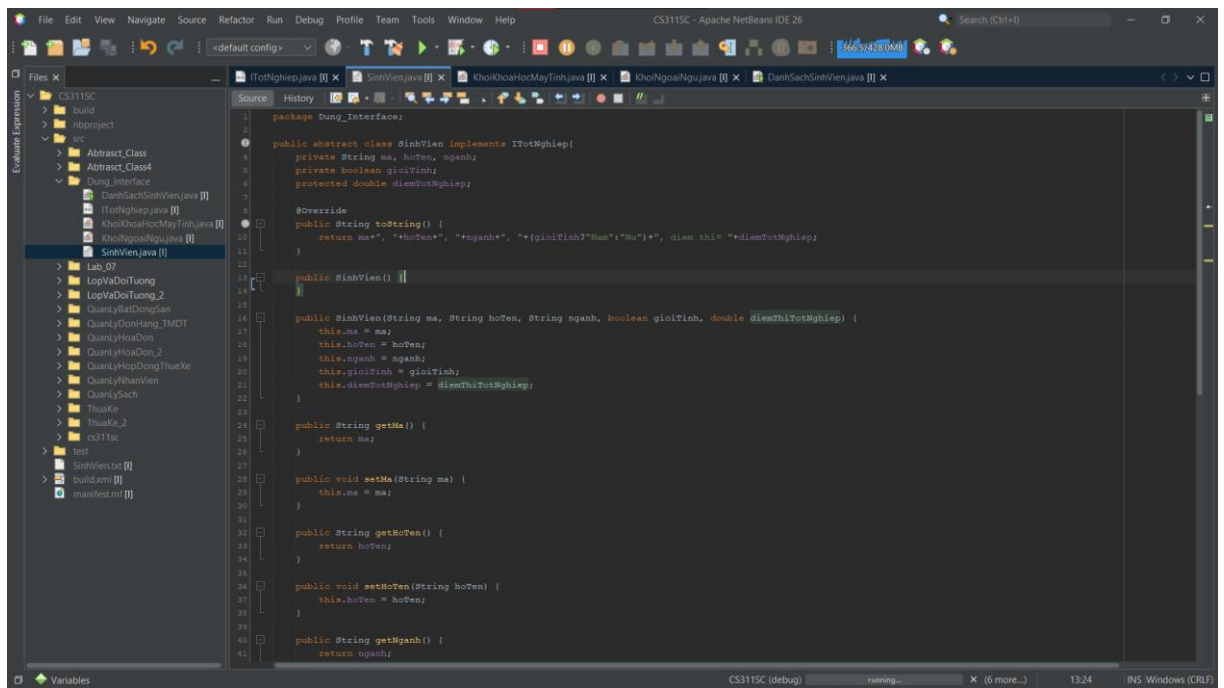
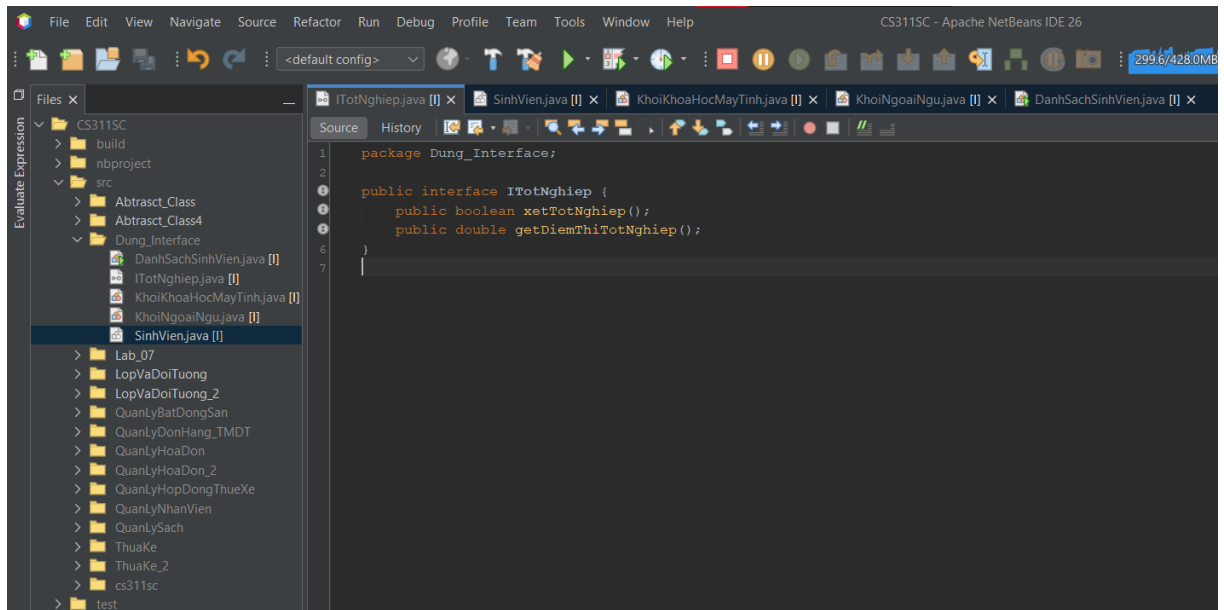
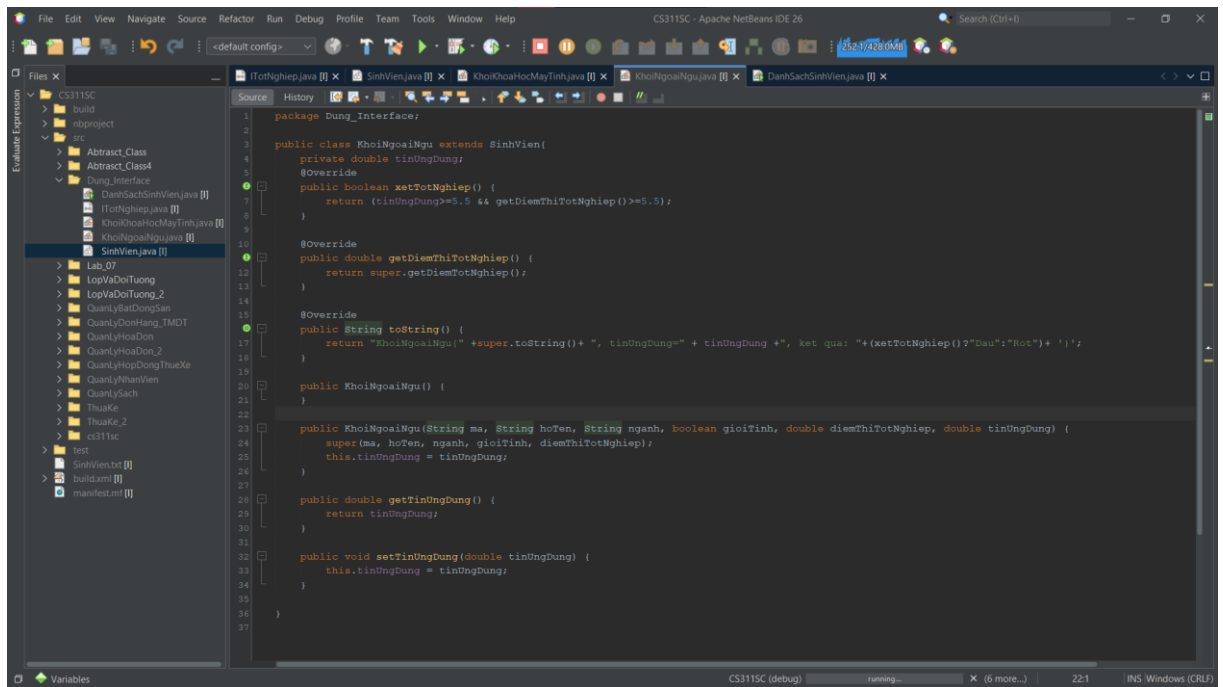
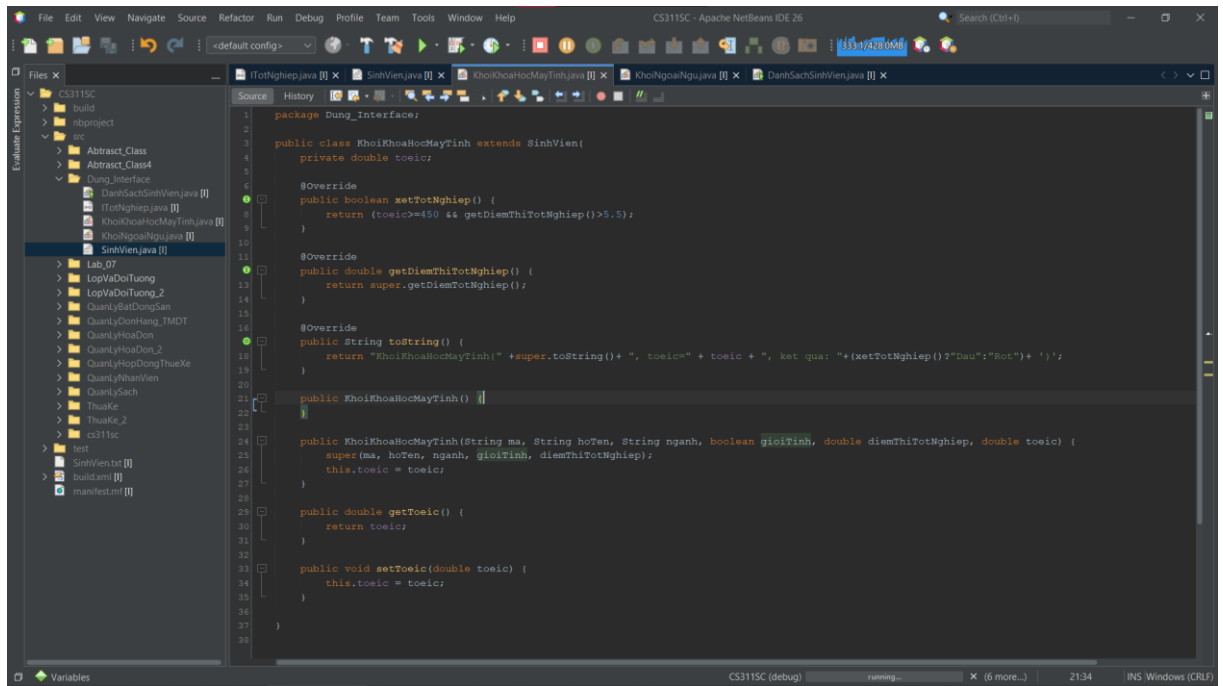






The screenshot shows the Apache NetBeans IDE with the project 'CS311SC' open. The 'DanhSachSinhVien.java' file is selected in the 'Files' pane. The 'Source' view displays the following code:

```
1 package Dung_Interface;
2
3 import java.io.File;
4 import java.io.PrintStream;
5 import java.util.ArrayList;
6 import java.util.Scanner;
7
8 public class DanhSachSinhVien {
9     private ArrayList<ITotNghiep> list = new ArrayList<>();
10    public void docFile(String tenFile){
11        try {
12            File f = new File(tenFile);
13            if(f.exists()){
14                System.out.println("ok File");
15                Scanner read = new Scanner(f);
16                while(read.hasNext()){
17                    String line = read.nextLine();
18                    String s[] = line.split(",");
19                    for(int i=0;i<s.length;i++) s[i]=s[i].trim();
20                    boolean gioiTinh = Boolean.parseBoolean(s[3]);
21                    double diemTotNghiep = Double.parseDouble(s[4]);
22                    if(s[2].equalsIgnoreCase("khmt")) {
23                        double toaic = Double.parseDouble(s[5]);
24                        a.add(new KhoiKhoahocMayTinh(s[0], s[1], s[2], gioiTinh, diemTotNghiep, toaic));
25                    }else{
26                        double tinUngDung = Double.parseDouble(s[5]);
27                        a.add(new KhoiNgoingu(s[0], s[1], s[2], gioiTinh, diemTotNghiep, tinUngDung));
28                    }
29                }
30            }else System.out.println("ko co file");
31        } catch (Exception e) {
32        }
33    }
34
35    public void ghiFile(String tenFile){
36        try {
37            File f = new File(tenFile);
38            if(f.exists()){
39                System.out.println("File da ton tai");
40            }
41        }
42    }
43 }
```

The screenshot shows the Apache NetBeans IDE with the project 'CS311SC' open. The 'DanhSachSinhVien.java' file is selected in the 'Files' pane. The 'Source' view displays the following code:

```
34 }
35
36 public void ghiFile(String tenFile){
37     try {
38         File f = new File(tenFile);
39         if(f.exists()){
40             PrintStream out = new PrintStream(f);
41             for(ITotNghiep i:a){
42                 if(i.xetTotNghiep())out.println(i);
43             }
44             out.close();
45             System.out.println("ghi file thanh cong");
46         }else System.out.println("File da ton tai khong the ghi ");
47     } catch (Exception e) {
48     }
49 }
50
51 public void lietKe(ArrayList<ITotNghiep> b, String tieude){
52     System.out.println(tieude);
53     for(ITotNghiep i:b){
54         System.out.println("\t"+i);
55     }
56 }
57
58 public ArrayList<ITotNghiep> danhSachTotNghiep(){
59     ArrayList<ITotNghiep> b = new ArrayList<>();
60     for(ITotNghiep i:a){
61         if(i.xetTotNghiep()) b.add(i);
62     }
63     return b;
64 }
65
66 public double tinhDiemThiTotNghiepTrungBinh(){
67     double tong = 0;
68     for(ITotNghiep i:a){
69         tong+=i.getDiemThiTotNghiep();
70     }
71     return tong/a.size();
72 }
```

